EECE2311 - Summer 2 2025 Semester

Lab for EECE2310: Introduction to Digital Design and Computer Architecture

Lab Assignment 1

EECE2311 Lab Section 01

EECE2310 Course Section 01

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Abstract

This lab introduced schematic design using Quartus tools and validated the functionality of binary adders through software simulations and FPGA hardware implementation on a DE1-SoC board. Logical operations were effectively verified, demonstrating the application of binary addition using hardware switches and LEDs as visual indicators of circuit outputs.

Introduction

This lab exercise was designed to familiarize students with Quartus Prime Lite schematic software and FPGA-based hardware validation on the DE1-SoC board, specifically through the construction and testing of binary adders like the half adder and full adder. The primary objective was to verify circuit operation through Quartus simulations, followed by physical validation using FPGA hardware. Specific constraints included using only permitted schematic modules, precise FPGA pin assignments to ensure correct circuit functionality, and verifying circuits within a defined simulation interval (e.g., 0 to 200 ns). It was anticipated that the half adder circuit would correctly produce sum and carry outputs for all two-bit input combinations (00, 01, 10, 11), and the full adder that consists of two half adders would accurately handle all three-bit input scenarios, confirmed by both simulations and LED indicators on the FPGA board.

Lab Setup

Pre-Lab

Half Adder Circuit Truth Table

A	*B*	*S*	C_{out}
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Full Adder Circuit Truth Table

X	Y	Z	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Equipment

- **Quartus Prime Lite Software**: used for designing, simulating, and validating digital circuit schematics before FPGA hardware implementation.

- **<u>DE1-SoC FPGA board</u>**: a hardware platform containing the Cyclone V FPGA chip used to physically implement and validate the binary adder circuits.
- **<u>USB-B to USB-A Cable</u>:** Used to connect the DE1-SoC FPGA board to the lab PC for programming and data transfer.
- **<u>Power Adapter:</u>** Power the DE1-SoC FPGA board.
- **<u>Lab PC:</u>** Run Quartus software and program the board.

Results and Analysis

Results

<u>Half Adder Circuit</u>

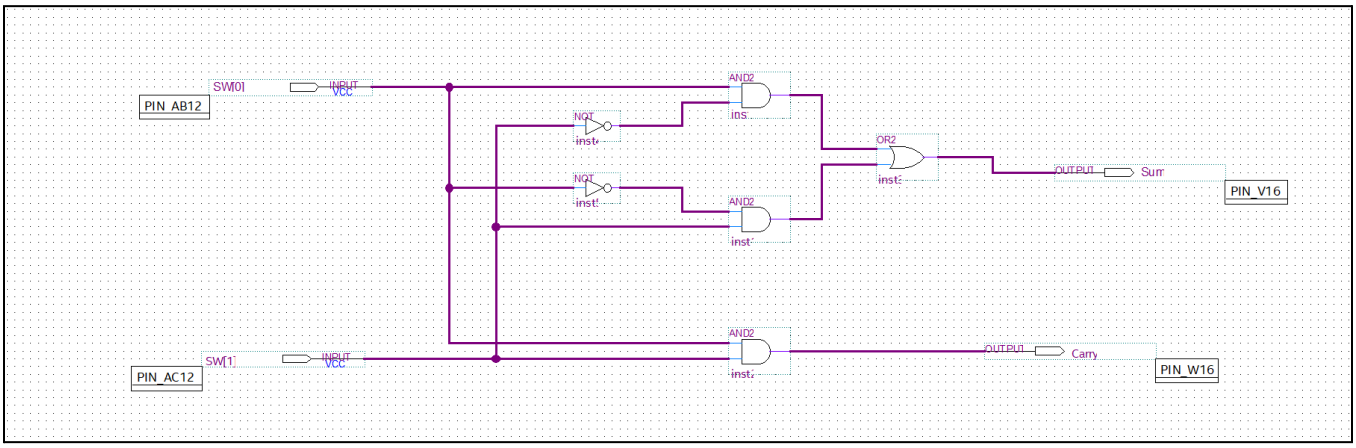


Figure 1: Schematic diagram of the half adder circuit, showing inputs SW[0] (PIN_AB12), SW[1] (PIN_AC12), and outputs Sum (PIN_V16), Carry (PIN_W16). SW refers to switch.

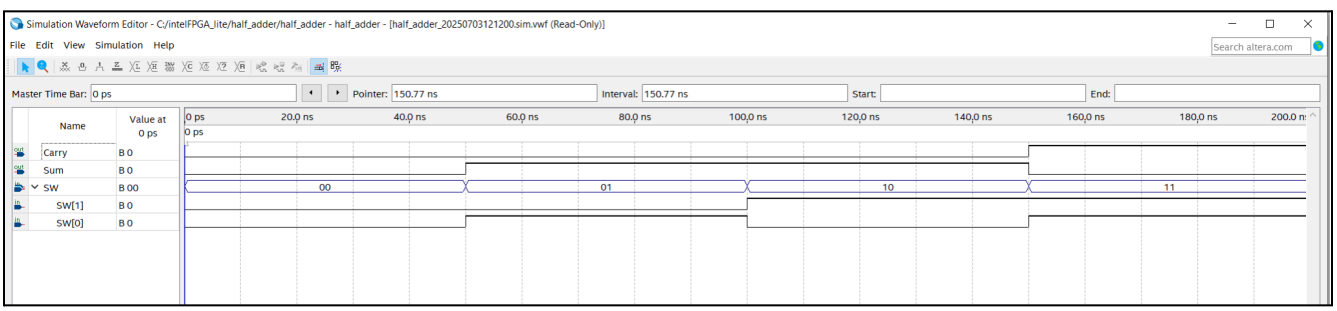


Figure 2**:** Simulation waveform showing the half adder functionality for all input combinations (00, 01, 10, 11). The outputs correctly show Sum as logic high (1) for inputs 01 and 10, and Carry as logic high (1) only for input 11.

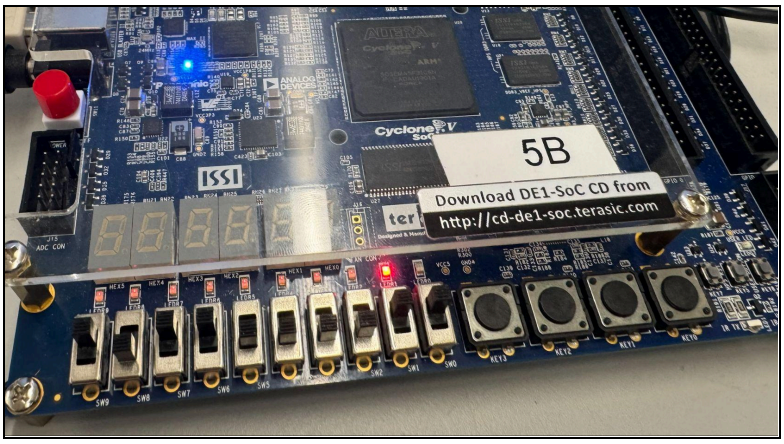


Figure 3: DE1-SoC FPGA board with SW[0] and SW[1] set high, illuminating red LEDR1 (Carry) of 1.

Full Adder Circuit

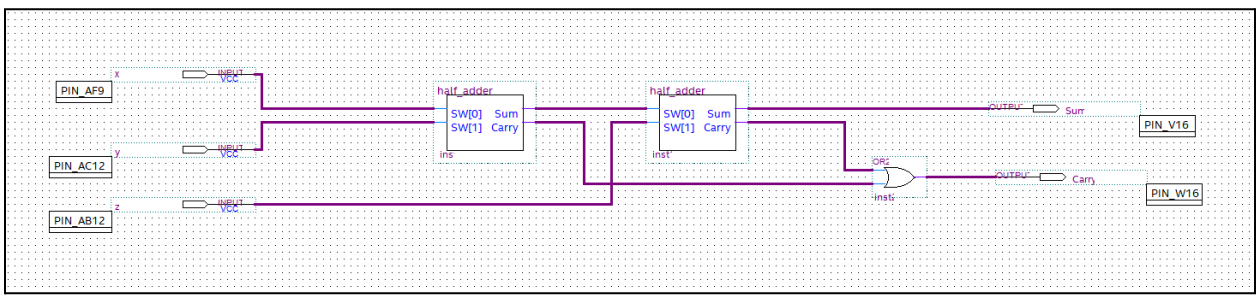


Figure 4: Schematic diagram of the full adder circuit, showing three inputs x (PIN_AF9), y (PIN_AC12), and z (PIN_AB12) and producing outputs Sum (PIN_V16) and Carry (PIN_W16).

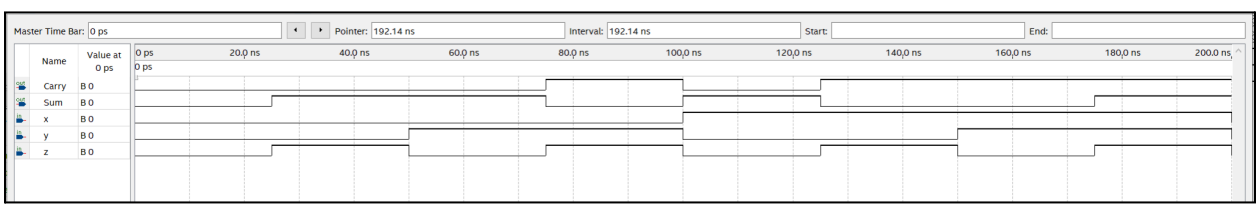
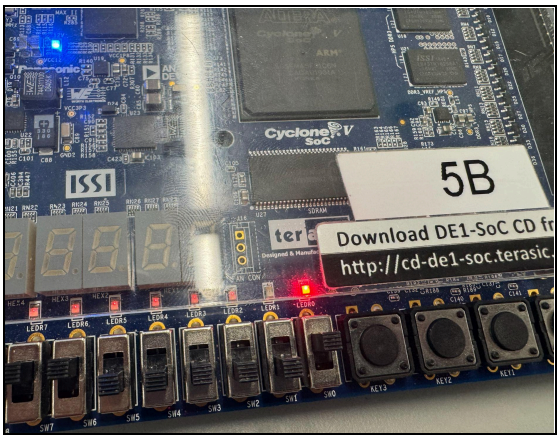


Figure 5: Simulation waveform verifying full adder functionality for all input combinations (000 through 111). Outputs correctly indicate Sum as high for inputs 001, 010, 100, and 111, and Carry as high for inputs 011, 101, 110, and 111.



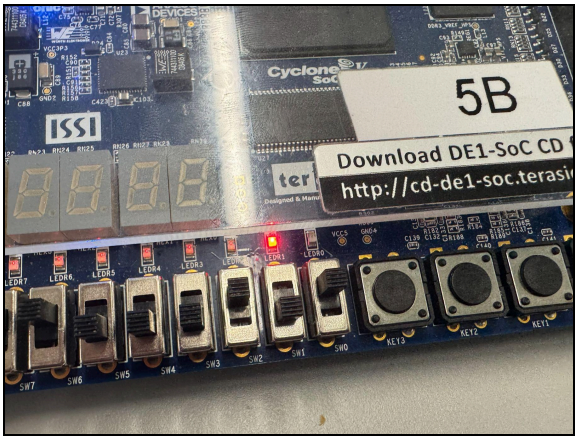
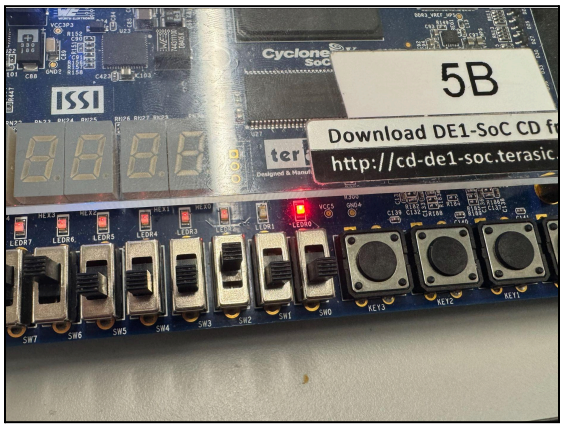


Figure 6 (left): DE1-SoC board showing input 100 with Sum=1, Carry=0.

Figure 7 (right): DE1-SoC board showing input 101 with Sum=0, Carry=1.



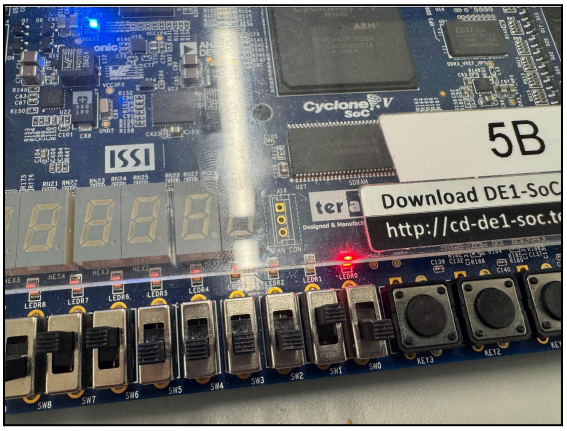


Figure 8 (left): DE1-SoC board showing input 001 with Sum=1, Carry=0.

Figure 9 (right): DE1-SoC board showing input 010 with Sum=1, Carry=0.

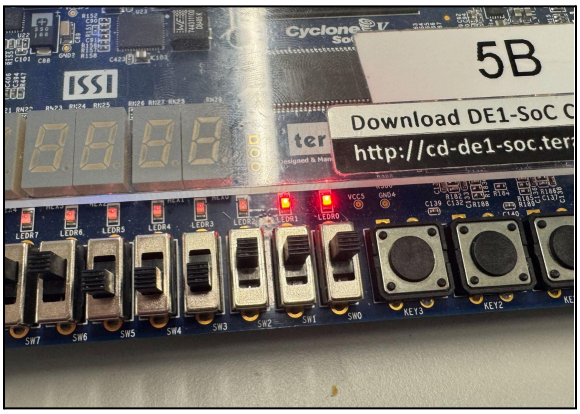


Figure 10: DE1-SoC board showing input 111 with Sum=1, Carry=1.

Analysis

Half Adder Circuit

The half adder simulation and hardware validation matched theoretical expectations precisely. The Sum output LED (LEDR0) correctly illuminated for input combinations 01 and 10, and the Carry output LED (LEDR1) correctly activated only for input combination 11 (Figure 3). Neither LED lights up for the input combination of 00. These results verified proper circuit operation and alignment with theoretical binary addition.

Full Adder Circuit

The full adder, constructed by two half adders and an OR gate, similarly produced results aligning with the theoretical predictions. For instance, Figure 10 confirms correct full adder operation for input 111, illuminating both the Sum and Carry indicators. Other input combinations (100, 101, 001, and 010) activated either the Sum or Carry LEDs as expected (Figure 6-9). Initially, however, a mix-up in FPGA pin assignments for two of the three inputs caused the LEDs to fail to illuminate. Once corrected based on the lab guide's pin assignment tables [1], the circuit functioned as intended.

Conclusion

The experiments conducted through this lab validated that the schematic designs and the simulations performed in Quartus Prime Lite reliably predict the actual FPGA hardware behavior. The successful implementation of both half and full adder circuits showcased the importance of precise schematic design and careful simulation before hardware deployment. The initial pin assignment oversight emphasized the need for precise attention to hardware configuration details. Future work that would be beneficial includes exploring more complex arithmetic circuits or integrating the adders into larger computational systems.

References

[1] Emad Aboelela, "Lab 1: Introduction to Quartus Schematic Binary Adders Using Simulation & the DE1-SoC Board", Northeastern University, July 2025.